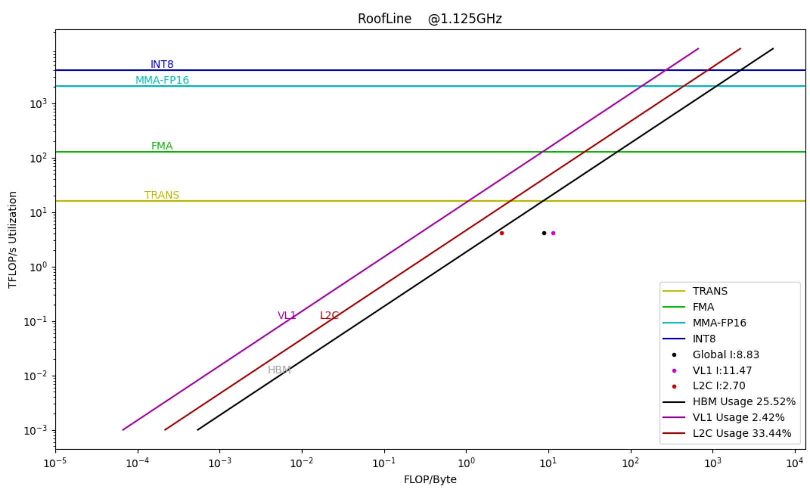
Sub-module: Summary

Name	Description	Value
Total Instructions	number of all instructions	20,773,993,970
Compute Instructions	number of compute instructions	18,212,683,209
Memory Instructions	number of memory instructions	2,561,310,761
Total Cycles	cycles use by kernel	9,756,915.91(Kcycles)
Memory Access per Second	memory access per second	358,868.57(MB/s)
Average AP busy cycles	Average busy cycles per AP	534,892,013.0
AP busy Duty	average AP busy duty of total cycles	5.48%
Compute Instructions busy Duty	Compute Instructions busy duty of total AP cycles	37.40%

Name: RoofLine

Description:



Sub-module: CE Statistics

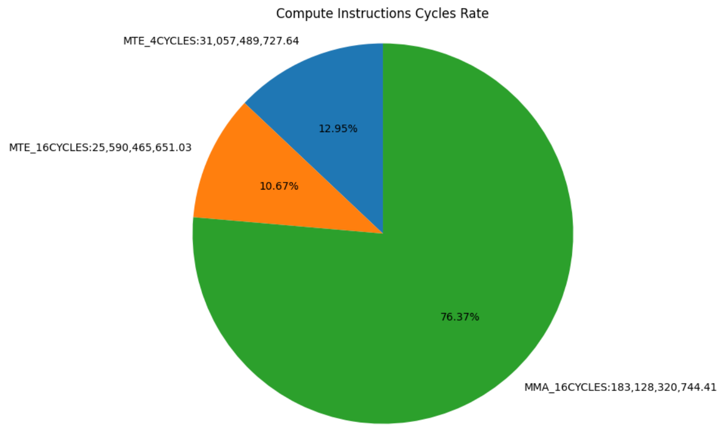
Name	Description	Value
WORKGROUPS	ce send workgroup	13,261,989.0
WAVES	ce send waves	27,177,390.0

Average Wave life cycles	average life cycles per wave from start to end	22,240.79

Sub-module: ISU Statistics

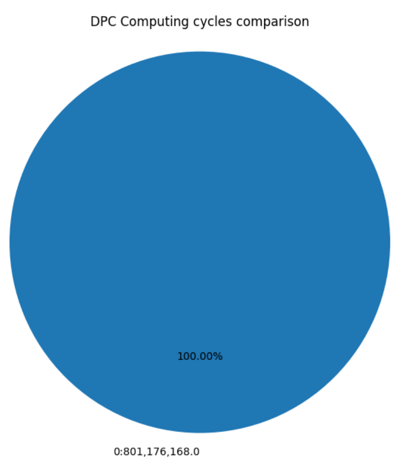
Name: Compute Instructions Cycles Rate

Description: rate between compute instructions cycles



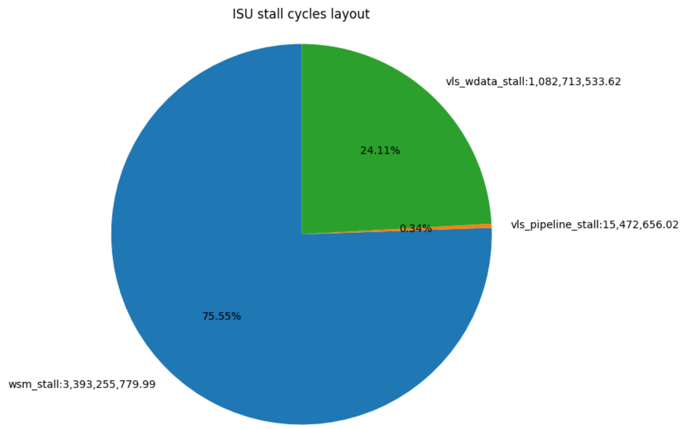
Name: DPC Computing cycles comparison

Description: Comparison on AP computing cycles



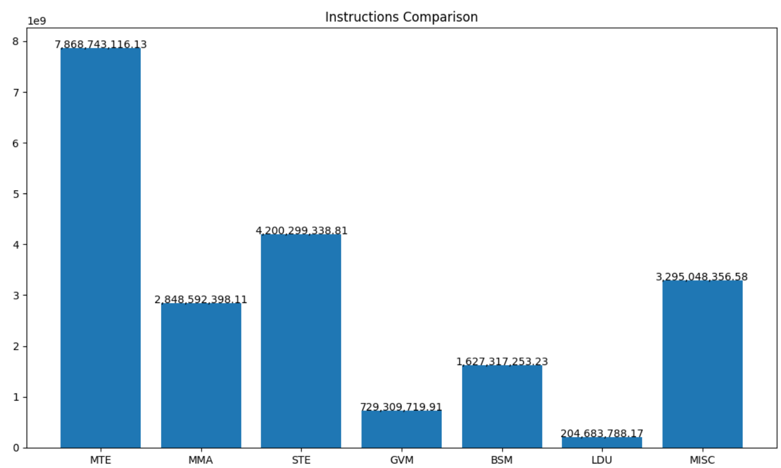
Name: ISU stall cycles layout

Description: ISU stall cycles layout



Name: Instructions Comparison

Description: Comparison between all instructions



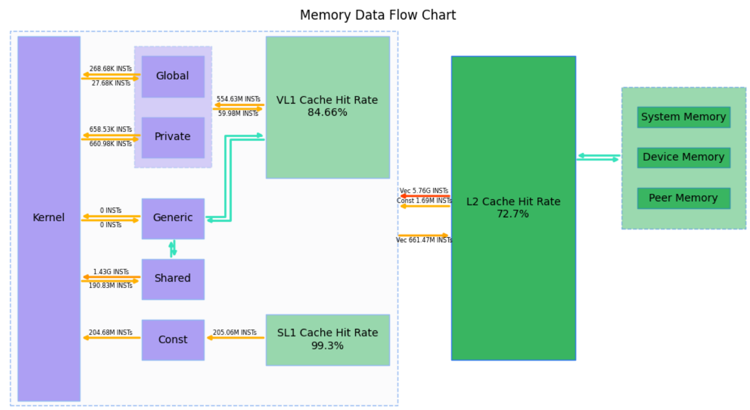
Sub-module: Memory Statistics

Name	Description	Value

Global Read Instructions	Number of global read instructions in all APs	268,682
Global Write Instructions	Number of global write instructions in all APs	27,683
Global Atomic Instructions	Number of global atomic instructions in all APs	60,011.43
Private Read Instructions	Number of private read instructions in all APs	658,534
Private Write Instructions	Number of private write instructions in all APs	660,975
Constant Read Instructions	Number of smem instructions in all APs	204,683,788
Constant Read SL1 Instructions	Num of LDU instructions in all SL1s	205,061,414
Constant Read L2 Instructions	Num of SL1 data miss requests	1,687,739
Constant SL1 Hit Rate	Total LDU instructions hit rate in all SL1s	99.30%
VL1 Read Instructions	total read instructions in all VL1s	554,634,888
VL1 Write Instructions	total write instructions in all VL1s	59,979,274
VL1 Atomic Instructions	total atomic instructions in all VL1s	117,989,851.60
VL1 Hit Rate	hit rate of all instructions in all VL1s	84.66%
L2C Read Instructions	total read instructions in all L2Cs	5,756,526,912
L2C Write Instructions	total write instructions in all L2Cs	661,466,976
L2C Atomic Instructions	total atomic instructions in all L2Cs	234,880,000.0
L2C Hit Rate	hit rate of all instructions in all L2Cs	72.70%
Total Atomic Instructions	total atomic instructions in memory access, include global atomic and shared memory atomic	118,516,624.71
Dnoc Read Req	number of dnoc read requests	1,454,283,392.0
Dnoc Write Req	number of dnoc write requests	322,270,368.0
Global Memory Read bytes	bytes read from global memory	182,395,745,280.0byte
Global Memory Write bytes	bytes write from global memory	41,249,123,328.0byte
Dnoc Read Average Latency	average latency of dnoc read request	277.08

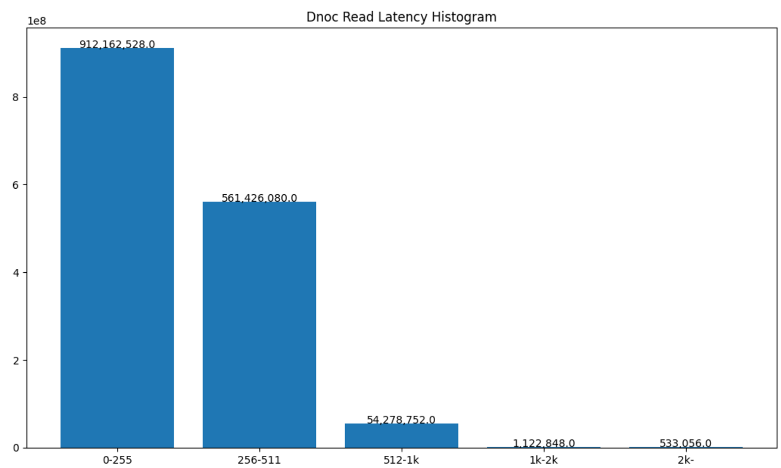
Name: Memory Data Flow Chart

Description:



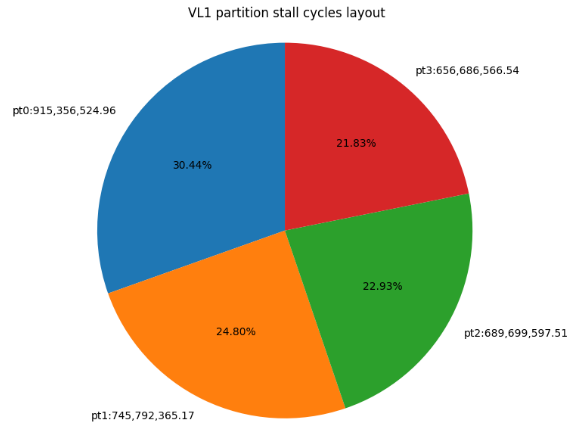
Name: Dnoc Read Latency Histogram

Description: histogram of dnoc read latency



Name: VL1 partition stall cycles layout

Description: VL1 partition access stall cycles layout



Sub-module: Workgroup Memory

Name	Description	Value
load instructions	number of total workgroup memory load instructions	1,431,701,323.27
store instructions	number of total workgroup memory write instructions	190,834,030.96
atomic instructions	number of total workgroup memory atomic instructions	526,773.11
total instructions	number of total workgroup memory instructions	1,627,317,253.23
average latency per load instruction	average latency per WG-load from issue to mem-ops done	0.04
average latency per store instruction	average latency per WG-store from issue to memops done	56.70
average latency per atomic instruction	average latency per WG-atom from issue to memops done	105.15
average latency of all stages per instruction	average latency of all WG-inst from issue to memops done	12.46
average cycles per load instruction	average cycles per WG-load when issuing (with bank conflict)	8.02
average cycles per store instruction	average cycles per WG-store when issuing (with bank conflict)	0.00
average cycles per atomic instruction	average cycles per WG-atom when issuing (with bank conflict)	52.22

average conflict cycles per instruction	average conflict cycles per WG-inst when bank conflict happens	3.52
average cycles per instruction	average cycles per WG-inst when issuing (with bank conflict)	7.07
shared memory access efficiency	Proportion of NON-CONFLICT access	50.19%

Sub-module: Occupancy

Name	Description	Value
Achieved waves	number of achieved waves	27,177,390.00
Dispatched waves	number of dispatched waves	27,177,288.0

Sub-module: GPU Throughput Statistics

Name	Description	Value
AP active cycles	number of wave instruction cycles executed on the AP	55,594,317,202.98
AP MTE Duty ratio	MTE Duty ratio relative to AP active	13.97%
AP STE Duty ratio	STE Duty ratio relative to AP active	7.56%
AP MMA Duty ratio	MMA Duty ratio relative to AP active	20.56%
VLS Duty ratio	VLS Duty ratio relative to AP active	0.0%
L2C Duty ratio	L2C Duty ratio relative to L2C active	2.13%

Sub-module: Compute workload

Name	Description	Value
instruction per cycle	The number of instructions per busy cycle across all sub-unit instances	38.84
instruction throughput	The percent of achieved instructions and theoretical instructions during all busy cycles	0.37
instruction throughput efficiency	The percentage of the maximum instruction throughput achieved during all busy cycles	9.34%

Sub-module: Instruction Statistics

Name	Description	Value
instruction number	The total number of instructions	20,773,993,970.95
instructions per AP	The number of instructions per PEU	199,749,942.03